

Data Analytics

Professional Program – 240 Hours

Course Description

This course is designed to equip learners with strong skills in data analytics, focusing on data handling, visualization, business intelligence, and applied statistics. Learners will work with structured and unstructured data using SQL, Excel, Python, and visualization tools. The program also introduces machine learning for analytics and covers data storytelling, dashboards, and decision-making. By the end of the course, participants will be able to extract, transform, and analyze data, generate insights, build interactive reports, and support data-driven business strategies.

Module 1: Foundations of Data Analytics

Syllabus Topics

- Introduction to Data Analytics & Lifecycle
- Types of Data: Structured, Semi-structured, Unstructured
- Analytics Types: Descriptive, Diagnostic, Predictive, Prescriptive
- Data Ethics & Governance

Practicals

- Case study on descriptive vs predictive analytics
- Data classification (structured/unstructured examples)

Tools

- Excel
- Google Sheets

Module 2: Excel & Advanced Excel for Analytics

Syllabus Topics

- Basic Functions (SUM, IF, VLOOKUP, HLOOKUP, INDEX-MATCH)
- Pivot Tables & Pivot Charts
- Conditional Formatting & Data Validation
- Advanced Functions (Power Query, Power Pivot)
- Macros & Automation Basics

Practicals

- Create sales reports with Pivot Tables
- Automate repetitive tasks with macros

- Data cleaning with Power Query

Tools

- Microsoft Excel
 - Google Sheets
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Module 3: SQL for Data Analytics

Syllabus Topics

- Database Basics & ER Models
- SQL Queries: SELECT, WHERE, GROUP BY, HAVING
- Joins (INNER, LEFT, RIGHT, FULL)
- Window Functions (RANK, LEAD, LAG)
- Subqueries & CTEs
- Data Extraction for Reporting

Practicals

- Write queries for business cases (e.g., top 5 products by revenue)
- Analyze sales and customer trends with SQL joins
- Create SQL-based dashboards with Power BI/Tableau

Tools

- MySQL
 - PostgreSQL
 - SQLite
 - SQLAlchemy
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Module 4: Data Visualization & Storytelling

Syllabus Topics

- Principles of Data Visualization
- Visual Encoding (Charts, Maps, Heatmaps)
- Storytelling with Data
- Choosing the Right Chart
- Creating Dashboards

Practicals

- Build dashboards for finance/sales datasets
- Create executive reports with KPIs

- Storytelling project: Present insights from dataset

Tools

- Tableau
 - Power BI
 - Plotly
 - Seaborn
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Module 5: Statistics for Data Analytics

Syllabus Topics

- Descriptive Statistics
- Probability Basics
- Inferential Statistics (Hypothesis Testing, T-tests, ANOVA)
- Correlation & Regression Basics
- Sampling & Bias in Data

Practicals

- Hypothesis testing with real datasets
- Correlation analysis between sales & marketing spend
- Regression model for predicting revenue

Tools

- Excel
- Python (SciPy, StatsModels)

- R

Module 6: Python for Data Analytics

Syllabus Topics

- Python Basics: Variables, Loops, Functions
- Data Analysis Libraries (NumPy, Pandas)
- Data Cleaning (Missing Values, Outliers, Transformations)
- Exploratory Data Analysis (EDA)
- Automating Reports with Python

Practicals

- Perform EDA on datasets (finance, healthcare, retail)
- Automate report generation with Pandas

- Merge multiple data sources with Python

Tools

- Python
 - Jupyter
 - Pandas
 - NumPy
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Module 7: Business Intelligence (BI) Tools

Syllabus Topics

- BI Concepts & Architecture
- Power BI Dashboard Creation
- Tableau Advanced Features
- Integrating SQL with BI tools
- Data Refresh & Scheduling

Practicals

- Build interactive BI dashboards
- Connect Power BI with SQL database
- Publish and share dashboards

Tools

- Power BI
 - Tableau
 - Google Data Studio
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Module 8: Machine Learning for Analytics

Syllabus Topics

- Introduction to Machine Learning in Analytics
- Supervised Learning: Regression, Classification
- Unsupervised Learning: Clustering
- Predictive Analytics with ML
- Model Evaluation (Confusion Matrix, Accuracy, RMSE)

Practicals

- Customer churn prediction (Classification)
- Sales forecasting (Regression)

- Customer segmentation (Clustering)

Tools

- Scikit-learn
 - Python
 - Jupyter
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Module 9: Big Data & Cloud Analytics

Syllabus Topics

- Introduction to Big Data & Analytics Ecosystem
- Hadoop & Spark Basics
- Data Warehousing Concepts
- Cloud Platforms for Analytics (AWS, Azure, GCP)
- Real-time Analytics (Kafka, Spark Streaming)

Practicals

- Query big datasets with Spark SQL
- Analyze data in AWS S3/Google BigQuery
- Stream processing example with Kafka

Tools

- Hadoop
 - PySpark
 - AWS S3
 - BigQuery
 - Databricks
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Module 10: Capstone Project & Soft Skills

Syllabus Topics

- End-to-End Analytics Project Lifecycle
- Business Problem Framing & Solution Design
- Communication & Data Storytelling
- Soft Skills: Presentation, Reporting, Consulting Approach
- Capstone Project Examples: Sales forecasting, Customer segmentation, Financial risk analysis

Practicals

- Work on real datasets (finance, healthcare, retail, marketing)

- Build BI dashboards + predictive models
- Present business insights to stakeholders

Tools

- Excel
 - SQL
 - Python
 - Power BI/Tableau
 - Cloud Platforms
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Program Outcomes

- Ability to collect, clean, preprocess, and integrate data from multiple sources for effective analysis.
- Apply statistical, mathematical, and analytical techniques to interpret data and identify patterns and trends.
- Use programming languages such as Python/R, SQL, and modern analytics tools to perform data analysis and automation tasks.
- Create meaningful data visualizations, dashboards, and reports to communicate insights clearly for decision-making.
- Build, evaluate, and apply basic predictive and machine learning models to solve real-world data analytics problems.
- Demonstrate ethical data handling, business understanding, teamwork, and adaptability to emerging data analytics technologies